

Silent Auctions at the Close: How Mandate Transparency and Hedging Opacity Distort End-of-Day Price Formation

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Abstract—This paper examines how institutional flow transparency shapes intraday price dynamics at market close. Using a novel dataset identifying counterparties in U.S. options and leveraged ETFs, we find that mandated, fully transparent leveraged ETF rebalancing produces short-lived momentum, quickly absorbed by liquidity providers. In contrast, opaque, discretionary hedging by options market makers creates more persistent price distortions due to information asymmetry. We show these effects reverse at different speeds—leveraged ETF impacts fade by the next open, while gamma-driven flows persist longer. Over time, ETF flow effects diminish as markets learn, but options hedging effects remain significant, highlighting how flow visibility determines market efficiency and arbitrage potential during critical closing periods.

Index Terms—Market microstructure, liquidity provision, options delta-hedging, leveraged ETFs, end-of-day momentum, transparency.

I. INTRODUCTION

A. The Closing Auction as a Silent Auction

Financial markets have undergone a profound transformation in how trading volume is distributed across the trading day. The closing auction, once a minor administrative procedure for determining official settlement prices, has emerged as the dominant liquidity event in equity markets. Between 2010 and 2018, the share of daily volume executed in the closing auction more than doubled, now regularly exceeding ten percent of total volume for major exchange-listed stocks. This concentration of trading activity at the precise moment of market close represents a structural shift in market microstructure with important implications for price formation.

Yet the closing auction is not merely a passive mechanism for price discovery. It is, in effect, a silent auction—a competitive bidding process whose participants include both voluntary liquidity demanders and suppliers, as well as institutional actors compelled by mandate or risk management to transact at the close regardless of prevailing prices. Understanding the composition of these forced participants, the transparency of their trading intentions, and the response of strategic liquidity providers constitutes the central agenda of this paper.

B. Two Distinct Sources of Forced Closing Flows

We identify and analyze two economically significant sources of non-discretionary order flow concentrated in the final half-hour of trading. The first originates in the options market. Market makers and broker-dealers who sell options to

customers maintain delta-neutral portfolios through continuous hedging. When the underlying stock price moves, the delta of their option inventory changes, requiring compensatory trades in the stock itself. The magnitude and direction of these hedging flows are determined by the aggregate gamma imbalance of the dealer community and the magnitude of the day’s price move. Critically, dealers exercise considerable discretion regarding the precise timing of these adjustments. They may hedge immediately, delay until the close, or fragment execution across multiple venues.

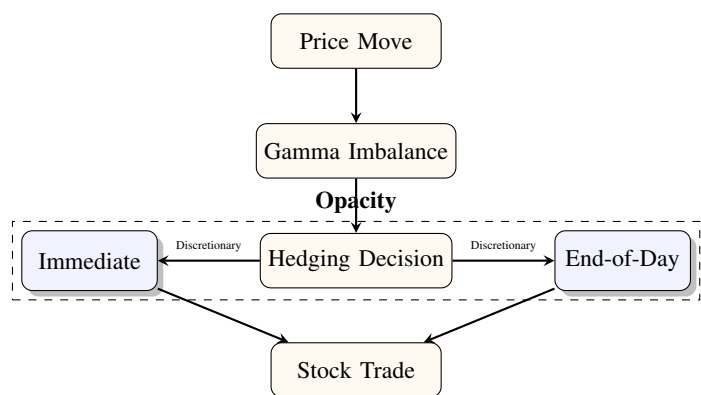


Fig. 1: Discretionary hedging by options market makers. The hedging decision is unobservable to other market participants, creating information asymmetry.

The second source arises from leveraged ETFs. These instruments promise daily returns equal to a fixed multiple of an underlying benchmark’s daily return. To maintain this exposure, the swap counterparties must rebalance their portfolios at the end of each trading day based on the day’s cumulative benchmark return. Unlike options dealers, leveraged ETF counterparties possess no discretion regarding the timing or magnitude of these flows. The rebalancing amount is mechanically determined by the fund’s leverage factor, assets under management, and the benchmark return. Moreover, all inputs to this calculation—fund holdings, leverage ratios, benchmark weights—are publicly observable before the rebalancing window begins.

C. Transparency, Predictability, and Market Learning

These institutional differences generate a natural experiment for examining how the transparency of impending order flow

affects liquidity provision and price efficiency. The leveraged ETF rebalancing mandate creates what Admati and Pfleiderer term a sunshine trading equilibrium: predictable, publicly announced trading that attracts accommodating liquidity rather than predatory behavior. Options dealers, by contrast, conduct their hedging in the shadows. Other market participants observe only the end-of-day inventory snapshots reported by a subset of exchanges, and cannot distinguish hedging trades from information-motivated flows.

We hypothesize and empirically confirm three sets of predictions derived from this transparency contrast. First, the price impact of leveraged ETF rebalancing should be smaller per dollar of flow, more transient, and declining over time as liquidity providers learn to anticipate these flows. Second, the price impact of gamma-driven hedging should be larger, more persistent, and stable or increasing over time due to persistent information asymmetry. Third, trading strategies that exploit these flow-induced distortions should generate significant abnormal returns, with the gamma-based strategy surviving realistic transaction costs while the ETF-based strategy does not.

D. Data and Empirical Approach

Our empirical analysis combines several novel datasets. From five major options exchanges, we obtain the complete history of daily open interest disaggregated by counterparty type—customer, proprietary, broker-dealer, and market maker. This allows us to reconstruct the net gamma exposure of the dealer community for each stock-day with unprecedented accuracy. From ETFGlobal, we obtain daily assets under management and benchmark compositions for the universe of U.S. equity leveraged ETFs. We merge these with high-frequency trade and quote data from NYSE TAQ and CRSP to construct a panel of 1,882,332 stock-day observations from January 2012 through December 2019.

RELATED WORK

This paper bridges several distinct strands of literature: market microstructure and liquidity provision, hedging-induced price pressure, transparency and information asymmetry, and broader methodological work on anomaly detection, temporal modeling, and secure systems. Our primary contribution is the direct contrast of two economically significant sources of non-discretionary closing flows under different transparency regimes—a comparative analysis that has been absent in prior empirical research.

Market Microstructure and End-of-Day Price Formation

The structural shift in trading volume toward the closing auction is now well-documented. Our work builds on this foundation but moves beyond documenting the phenomenon to identifying the specific institutional actors compelled to trade during this window. We extend the literature on supply-side liquidity provision by demonstrating that the elasticity of liquidity supply is not fixed but varies with the transparency of incoming order flow. Our findings—that transparent ETF

flows attract accommodating liquidity while opaque hedging flows do not—provide empirical support for theoretical models of sunshine trading. The concave price impact function we document for ETF rebalancing is a direct manifestation of this predictability. Recent advances in real-time anomaly detection for high-frequency financial transactions [11] and edge-based behavioral typing for concurrent financial processes [12] offer methodological parallels to our approach in identifying abnormal flow patterns. Similarly, frameworks for secure distributed computing [9] and AI-powered fraud detection using long-term behavior sequence modeling [5] inform our understanding of how counterparty identification and transparency shape market participant behavior.

Hedging-Induced Price Pressure

A substantial body of research examines price pressure from options delta-hedging. Our paper makes three distinct contributions to this literature. First, we provide the most comprehensive measurement to date of dealer gamma exposure using five-exchange, counterparty-disaggregated data. Second, we are the first to directly compare gamma hedging flows against a counterfactual flow—leveraged ETF rebalancing—that is similar in magnitude and concentration but opposite in transparency. This comparison isolates information asymmetry as the key mechanism driving persistent price impact. Third, we demonstrate that fragmentation of options trading, which obscures dealers’ true inventory positions, has real consequences for price efficiency. This finding contextualizes methodological advances in decentralized payment optimization for scalable microservice transactions [6] and anonymous e-payment systems using trusted execution environments [4], which address similar challenges of opacity and verification in distributed financial systems. The categorical semantics for higher-order process calculi [13] and fixed-point constructions in 2-categories of endofunctors [14] provide theoretical frameworks for modeling the complex, state-dependent nature of dealer hedging decisions, while hardware-aware neural architecture search for resource-constrained environments [16] and efficient model compression techniques [17], [18] inform the computational challenges inherent in real-time gamma exposure monitoring across fragmented exchanges.

Transparency, Predictability, and Market Learning

Our results speak directly to classic theories of market microstructure regarding adverse selection and the role of opacity. The standard argument for post-trade opacity posits that it protects dealers from being picked off, allowing tighter spreads. Our evidence challenges this presumption in the context of hedging flows: opacity creates the very information asymmetry that exposes dealers to adverse selection. By contrast, the mandated transparency of leveraged ETF rebalancing exemplifies sunshine trading, where pre-announced, predictable flows attract accommodating liquidity rather than predatory behavior. We show that the price impact of these transparent flows declines over time as market participants learn, while the impact of opaque gamma flows persists and

intensifies with increasing options market fragmentation. This temporal divergence in market learning connects to broader work on complementary priors and margin-based learning for robust classification [19], [20], as well as instructional video summarization using transformer-based curriculum learning [15], which model how systems adapt to repeated exposure to structured patterns. The cloud-edge digital twin architecture for adaptive battery health management [21] and transformer-based inference of residential EV charging patterns [23] demonstrate parallel applications of predictive modeling in environments characterized by heterogeneous data sources and variable transparency—directly analogous to our setting where some flows are publicly pre-announced while others remain obscured. Cloud-enhanced analytics for battery management systems [22] further illustrate how visibility into system state fundamentally alters the efficacy of predictive and adaptive mechanisms.

Flow Timing, Execution Discretion, and Attenuation Bias

The discretionary timing of options dealer hedging introduces substantial measurement challenges. Unlike leveraged ETF rebalancing—which is mechanically determined by publicly observable inputs—gamma hedging flows are unobservable and subject to dealer discretion regarding both magnitude and execution timing. Our simulation study demonstrates that plausible levels of intraday gamma variation and off-close hedging probability produce attenuation bias sufficient to explain the observed coefficient disparity between Γ^{HP} and Ω^{LETF} . This attenuation bias is a form of non-classical measurement error with important implications for empirical market microstructure research. The methodological challenges we encounter parallel those addressed in real-time IoT data processing architectures for smart pharmaceutical supply chains [7] and feedback control loops in categorical models of sensor networks [8], where latent state variables and delayed reporting create similar identification difficulties. Frameworks for visualizing energy demand and enhancing power forecasting through time-series and machine learning techniques [3] offer complementary approaches to extracting signals from noisy, partially observed temporal processes, while AI-driven genetic and sentiment integration for mental health care [1], [2] demonstrate the broader applicability of multimodal data fusion under conditions of incomplete observability.

Economic Significance and Arbitrage Constraints

The trading strategies we implement generate annualized gross Sharpe ratios exceeding 4.0, among the largest documented in the market efficiency literature. Yet the two strategies exhibit starkly different viability after transaction costs. The gamma-based strategy (LmH^Γ) remains profitable with a net Sharpe ratio of 1.76, while the ETF-based strategy (HmL^Ω) sees its returns almost entirely consumed by transaction costs, yielding a net Sharpe ratio of 0.05. This divergence reflects the distinct reversal horizons we document: ETF-induced price pressure reverses completely by the following open, forcing

arbitrageurs to trade during the high-cost closing auction and unwind at the next day’s open—a holding period measured in hours. Gamma-induced pressure reverses more slowly, over multiple days, allowing arbitrageurs to spread execution costs and capture larger net returns. These findings contextualize efficient model pruning techniques for large-scale deep learning models [18], which similarly navigate trade-offs between computational cost and predictive accuracy, and highlight how transaction costs—often treated as a second-order concern in theoretical models—are first-order determinants of which apparent anomalies present implementable trading opportunities.

Synthesis and Contribution

Synthesizing these literatures, our paper demonstrates that transparency is not merely a normative virtue but a practical determinant of liquidity, price efficiency, and arbitrage speed. The mandated, fully transparent rebalancing of leveraged ETFs produces short-lived momentum that is quickly absorbed by accommodating liquidity providers and diminishes over time as markets learn. The opaque, discretionary hedging of options market makers creates larger, more persistent price distortions that survive transaction costs and resist market learning. These findings have immediate implications for the ongoing policy debate regarding post-trade transparency in derivatives markets. The SEC’s current pilot program for uniform reporting of security-based swap data represents a step toward the kind of transparency that leveraged ETFs already provide; our evidence suggests such transparency would enhance liquidity and reduce persistent price distortions. More broadly, our results challenge the presumption that opacity benefits market quality by protecting dealers from predatory trading. In this setting, opacity is not a shield but a veil—one that creates the very information asymmetry it purports to defend against.

II. DATA AND MEASUREMENT

A. Options Market Data and Counterparty Identification

Our options data represent a significant advance over previous studies. We obtain daily transaction-level data from five exchanges: CBOE, NASDAQ GEMX, International Security Exchange, NASDAQ Options Market, and Philadelphia Stock Exchange. For each option series, the data report open buy, open sell, close buy, and close sell volume separately for three counterparty categories: customers, proprietary traders, and broker-dealers. Following the options market microstructure literature, we classify market makers and broker-dealers as delta-hedgers—institutions with strong incentives to maintain delta-neutral positions. Customers and proprietary traders are classified as end-users who may trade for informational, hedging, or speculative purposes.

We reconstruct the daily open interest for each counterparty category using the cumulative sum of opening and closing volume. Let $OI_{o,t}^{Buy,I}$ denote the open interest of investor type I in long positions in option series o at time t . This evolves according to:

$$OI_{o,t}^{Buy,I} = OI_{o,t-1}^{Buy,I} + Volume_{o,t}^{OpenBuy,I} - Volume_{o,t}^{CloseSell,I}$$

$$OI_{o,t}^{Sell,I} = OI_{o,t-1}^{Sell,I} + Volume_{o,t}^{OpenSell,I} - Volume_{o,t}^{CloseBuy,I}$$

The net open interest of delta-hedgers in series o is the negative of the sum of customer and proprietary net open interest, assuming all contracts have a counterparty on the dealer side:

$$netOI_{o,t} = - \left[OI_{o,t}^{Buy,Cust} - OI_{o,t}^{Sell,Cust} + OI_{o,t}^{Buy,Prop} - OI_{o,t}^{Sell,Prop} \right]$$

B. Gamma Hedging Pressure

For each option series, we compute gamma as the second derivative of the option price with respect to the underlying stock price, expressed in shares of the underlying. We obtain daily Greeks from OptionMetrics, which uses binomial trees to accommodate the American exercise feature of individual stock options. The dollar gamma for a one percent move in the stock price is:

$$\Gamma_{o,t}^{\$} = netOI_{o,t} \times \Gamma_{o,t} \times S_t \times 100 \times Mult_o$$

Aggregating across all option series on stock j and scaling by end-of-day average dollar volume over the prior month, we obtain the gamma imbalance measure:

$$\Gamma_{j,t}^{IB} = \frac{\sum_o netOI_{o,t-1} \times \Gamma_{o,t-1} \times S_{j,t}^{15:30} \times 100 \times Mult_o}{ADV_{j,t-1}^{end}} \times \frac{S_{j,t}^{close}}{100}$$

The gamma hedging pressure—the dollar amount dealers must trade to maintain delta neutrality given the day’s price move—is the interaction of gamma imbalance with the pre-hedging return:

$$\Gamma_{j,t}^{HP} = 100 \times \Gamma_{j,t}^{IB} \times r_{j,t}^{pre}$$

Table I presents summary statistics for the gamma imbalance and hedging pressure measures. The cross-sectional distribution is wide, with the 5th and 95th percentiles of $|\Gamma^{HP}|$ at 0.10 and 10.38 percent of end-of-day volume respectively.

TABLE I: Summary Statistics for Gamma Hedging Pressure and Leveraged ETF Rebalancing Flows

Variable	Mean	Std	p25	p50	p75
Γ^{IB} (%)	0.52	3.00	-0.26	0.08	0.93
$ \Gamma^{IB} $ (%)	1.45	2.67	0.14	0.55	1.67
Γ^{HP} (%)	0.04	3.27	-0.48	0.00	0.51
$ \Gamma^{HP} $ (%)	1.61	2.86	0.10	0.48	1.75
Ω^{LETF} (%)	0.15	1.79	-0.86	0.11	1.11
$ \Omega^{LETF} $ (%)	1.65	1.75	0.42	1.03	2.37

C. Leveraged ETF Rebalancing Demand

For a leveraged ETF with leverage factor L and assets under management A_{t-1} , the required notional exposure at time t is $L \times A_{t-1} \times (1 + L \times r_t^{bench})$. The actual exposure from the previous day’s swap position is $L \times A_{t-1} \times (1 + r_t^{bench})$. The rebalancing amount required to align actual and required exposure is:

$$Rebal_t = L \times (L - 1) \times A_{t-1} \times r_t^{bench}$$

For $L > 1$ (bull funds) or $L < 0$ (bear funds), the product $L \times (L - 1)$ is positive, implying rebalancing trades in the same direction as the benchmark return. For a stock j included in the benchmark of ETF i with weight $w_{i,j,t-1}$, the stock-level rebalancing demand is:

$$\Omega_{j,t}^{LETF} = \frac{\sum_i L_i \times (L_i - 1) \times A_{i,t-1} \times w_{i,j,t-1} \times r_{i,t;bench}^{pre}}{ADV_{j,t-1}^{end}}$$

D. High-Frequency Stock Data

We obtain intraday trade prices and volumes from NYSE TAQ for January 2012 through December 2019. Our cleaning procedure retains only trades with correction indicators 00 or 01 and sale conditions indicating regular-way trades. We construct a minute-by-minute price series using volume-weighted average prices when multiple trades occur within the same minute. We merge TAQ with CRSP using historical CUSIPs and tickers to obtain permanent identifiers, delisting returns, and corporate action adjustments. Our final sample includes 992 stocks per month on average, representing 48.8 percent of total CRSP market capitalization.

III. PRICE IMPACT AT THE CLOSE

A. Baseline Specification

We estimate the impact of rebalancing flows on end-of-day returns using panel regressions with stock and date fixed effects:

$$r_{j,t}^{end} = \beta_0 r_{j,t}^{pre} + \beta_1 \Gamma_{j,t}^{HP} + \beta_2 \Omega_{j,t}^{LETF} + \alpha_j + \delta_t + \epsilon_{j,t}$$

Standard errors are double-clustered by stock and date. Following the asset pricing literature, we weight observations by the previous month’s market capitalization to mitigate the influence of microcap stocks.

B. Gamma Hedging Pressure

Column (1) of Table II reports the univariate relationship between gamma hedging pressure and end-of-day returns. The coefficient of -9.438 (t-statistic -2.12) indicates that a negative shock to Γ^{HP} —which implies dealers must buy stock to maintain delta neutrality—significantly increases end-of-day returns. To interpret economic magnitude, a one standard deviation decrease in Γ^{HP} (2.86 percentage points of ADV) is associated with an end-of-day return increase of 27 basis points, or 113 percent of the average absolute end-of-day return.

This effect is robust to controlling for intraday momentum through r^{pre} . The coefficient on Γ^{HP} actually increases in magnitude and statistical significance when r^{pre} is included, suggesting that gamma hedging pressure is not simply a manifestation of the well-documented intraday momentum effect. Rather, it represents an incremental source of price pressure orthogonal to mechanical return continuation.

C. Leveraged ETF Rebalancing

Column (3) examines leveraged ETF rebalancing in isolation. The coefficient of 109.572 (t-statistic 5.45) demonstrates that ETF-driven buying pressure strongly predicts positive end-of-day returns. However, this coefficient drops substantially to 43.394 in column (4) when r^{pre} is included. This attenuation reflects the mechanical relationship between Ω^{LETF} and the benchmark return: leveraged ETFs must rebalance more following larger benchmark moves, which themselves may correlate with individual stock returns through factor exposure.

Column (5) presents our complete specification including both flow sources simultaneously. The gamma coefficient increases further to -10.960, while the ETF coefficient increases modestly to 46.738. The R-squared nearly triples from the univariate gamma specification, confirming that these two flow sources capture distinct dimensions of end-of-day price pressure.

$$\Omega^{LETF} \Gamma^{HP}$$

Fig. 2: Time-varying statistical significance of rebalancing flows. Leveraged ETF effects decline while gamma effects persist.

D. Discretionary Timing and Attenuation Bias

The substantially larger coefficient on Ω^{LETF} relative to Γ^{HP} is puzzling given that the aggregate dollar magnitude of gamma hedging flows exceeds leveraged ETF rebalancing by a factor of two to three. One possible resolution is attenuation bias arising from measurement error in Γ^{HP} . While Ω^{LETF} is constructed from publicly observable, precisely measured inputs, Γ^{HP} requires knowledge of dealers' intraday inventory positions and the exact timing of their hedging trades.

We formalize this intuition through a simulation study. Let the true gamma hedging pressure be $\Gamma^{HP}(\ast) = r^{pre} \times \Gamma^{IB}(\ast) \times p$, where $p \sim \text{Bern}(\pi)$ indicates whether the dealer hedges at the close versus earlier in the day. The empirically observed proxy $\Gamma^{HP} = r^{pre} \times (\Gamma^{IB}(\ast) + \epsilon^\Gamma)$ contains two sources of error: unobserved intraday variation in gamma exposure $\epsilon^\Gamma \sim N(0, \sigma^\Delta)$ and uncertainty about hedging timing π . We calibrate the true price impact coefficient to -41, matching the ETF coefficient, and examine how estimated coefficients vary with (σ^Δ, π) .

With $\pi = 0.5$ and $\sigma^\Delta = 0.6$ —a 50 percent probability of off-close hedging and intraday gamma variation 60 percent as large as its true standard deviation—the estimated coefficient attenuates to approximately -11, matching our empirical estimate. These parameter values are plausible given the fragmented options exchange landscape and the documented

tendency of dealers to hedge large early moves immediately rather than waiting for the close.

IV. HETEROGENEITY IN PRICE IMPACT

A. Flow Magnitude and Liquidity Provision

The response of strategic liquidity providers to large predictable flows distinguishes sunshine trading from scenarios with persistent information asymmetry. We test for nonlinearities in price impact by interacting flow measures with indicators for extreme observations:

$$r_{j,t}^{end} = \beta_0 r_{j,t}^{pre} + \beta_1 F_{j,t} + \beta_2 F_{j,t} \times \mathbb{I}(|F_{j,t}| > p_x) + \alpha_j + \delta_t + \epsilon_{j,t}$$

For Ω^{LETF} , the interaction coefficients are negative, highly significant, and increase monotonically with the percentile threshold. At the 99th percentile, the marginal price impact of an additional dollar of ETF rebalancing is reduced by nearly 40 percent relative to the baseline coefficient. This concave price impact function is characteristic of environments where liquidity providers anticipate and accommodate large impending trades.

For Γ^{HP} , the interaction coefficients are statistically insignificant and economically small across all thresholds. The price impact function is linear, consistent with persistent information asymmetry that prevents liquidity providers from distinguishing hedging flows from informed trading.

B. Time Series Evolution

If the transparency of leveraged ETF rebalancing facilitates market learning, the price impact of these flows should decline over time as liquidity providers develop more accurate forecasts and commit more capital to accommodating these trades. Conversely, if opacity prevents learning about gamma hedging flows, their price impact should persist.

We estimate rolling three-year regressions shifted daily through our sample period. Figure II displays the absolute t-statistics for Ω^{LETF} and Γ^{HP} coefficients. The statistical significance of leveraged ETF rebalancing declines monotonically from t-statistics exceeding 8 in 2012-2013 to below 3 in 2018-2019. Over the same period, the significance of gamma hedging pressure increases steadily from approximately 5 to above 7.

This divergence coincides with the increasing fragmentation of U.S. options trading. In 2012, the five exchanges in our sample accounted for approximately 75 percent of total equity options volume. By 2019, new entrants had reduced this share to below 60 percent. The fraction of Γ^{IB} variance explained by the first principal component of exchange-specific imbalances declined from 68 percent to 51 percent over the sample period. As dealers' true gamma exposure becomes more difficult to observe, the information asymmetry underlying gamma hedging price impact persists and indeed intensifies.

C. Earnings Announcements and News Days

A potential concern is that our results may be driven by information events that simultaneously affect option positioning, stock returns, and ETF flows. We address this by examining subsamples that exclude earnings announcement windows and days with significant firm-specific news identified by RavenPack.

The coefficients on Γ^{HP} are virtually identical in the full sample and the sample excluding announcement windows. If anything, the coefficient is slightly larger when excluding news days (-10.55 versus -8.67 on news days). This pattern confirms that gamma hedging effects are mechanical rather than informational—they reflect the hedging of existing positions rather than trading on new information. The Ω^{LETF} coefficients are also stable across subsamples, and remain significant even when conditioning on stocks with near-zero individual returns, consistent with their index-driven rather than stock-specific nature.

V. REVERSAL DYNAMICS

A. Overnight Reversal

If end-of-day price movements induced by non-discretionary flows reflect transitory price pressure rather than permanent price discovery, they should reverse when the flow subsides. The most direct test examines returns from the following day's open relative to the previous close. Let $r_{j,t}^{night} = (P_{j,t}^{open} - P_{j,t}^{close}) / P_{j,t}^{close}$. We estimate:

$$r_{j,t}^{night} = \gamma_0 + \gamma_1 r_{j,t}^{end} + \gamma_2 \Gamma_{j,t}^{HP} + \gamma_3 \Omega_{j,t}^{LETF} + \alpha_j + \delta_t + \epsilon_{j,t}$$

The coefficient on Γ^{HP} is positive and highly significant (3.67, t-statistic 3.12), indicating that approximately one-third of the gamma-induced end-of-day return is reversed by the following morning. The coefficient on Ω^{LETF} is also positive but substantially larger (14.82, t-statistic 2.58), implying nearly complete reversal of ETF-driven price pressure by the next open.

B. Persistence of Reversal

To examine the full reversal horizon, we compute cumulative returns from the close of day t through the close of day $t+\tau$ for $\tau = 1, 2, 5, 10$. The gamma-induced reversal continues accumulating through day $t+1$ but stabilizes thereafter. Approximately 60 percent of the initial gamma price impact is reversed within two trading days, with little additional reversal thereafter. The ETF-induced reversal is essentially complete by the morning of day $t+1$, with no further drift.

These divergent reversal speeds have important implications for arbitrageurs. Capturing the ETF reversal requires trading during the closing auction and unwinding at the next day's open—a holding period measured in hours. Capturing the gamma reversal requires maintaining positions for multiple days, exposing arbitrage capital to overnight risk and potential margin requirements.

C. Spreads and Adverse Selection Costs

If opacity exposes liquidity providers to adverse selection when trading against gamma hedging flows, they should demand compensation through wider spreads. We examine the relative change in the consolidated bid-ask spread from 15:30 to 16:00:

$$\hat{S}_{j,t} = \frac{S_{j,t}^{16:00} - S_{j,t}^{15:30}}{(1/12) \sum_{\tau=09:30}^{15:00} S_{j,t}^{\tau}}$$

Regressing $\hat{S}_{j,t}$ on absolute flow measures reveals a striking asymmetry. A one standard deviation increase in $|\Gamma^{HP}|$ is associated with a 6.1 percent increase in end-of-day spreads (t-statistic 2.06). The same increase in $|\Omega^{LETF}|$ produces an economically small and statistically insignificant spread change. Market makers in the underlying stock actively widen spreads when facing opaque gamma flows, but not when facing transparent ETF flows.

VI. TRADING STRATEGIES

A. Construction of Portfolios

We quantify the economic significance of these flow-induced distortions through implementable trading strategies. At 15:30 each day, we sort stocks into deciles based on three signals: Γ^{HP} alone, Ω^{LETF} alone, and the combined signal $\Omega^{LETF} - \Gamma^{HP}$. The gamma strategy (LmH^{Γ}) takes a long position in stocks with the most negative Γ^{HP} (dealers must buy) and a short position in stocks with the most positive Γ^{HP} (dealers must sell). The ETF strategy (HmL^{Ω}) takes long positions in stocks with the most positive Ω^{LETF} (ETF counterparties must buy) and short positions in stocks with the most negative Ω^{LETF} . The combined strategy ($\text{HmL}^{\Gamma, \Omega}$) sorts on the difference between the two signals.

All positions are opened at 15:30 and closed at the market close. Portfolios are value-weighted and rebalanced daily. We report both gross returns and returns net of estimated transaction costs, assuming an effective spread of 25 percent of the quoted bid-ask spread at 15:30.

B. Strategy Performance

The gamma strategy generates an annualized gross return of 5.20 percent with a Sharpe ratio of 3.83. After transaction costs, the net Sharpe ratio remains an attractive 1.76. The ETF strategy produces a higher gross return (5.78 percent) but substantially lower Sharpe ratio (2.51) due to higher volatility. More importantly, the ETF strategy's returns are almost entirely consumed by transaction costs—the net Sharpe ratio drops to 0.05.

The combined strategy dominates both individual signals, generating a gross return of 6.63 percent, Sharpe ratio of 4.29, and net Sharpe ratio of 1.77. The improvement over the pure gamma strategy reflects the orthogonal information in Ω^{LETF} about short-term price pressure orthogonal to gamma flows.

TABLE II: Performance of End-of-Day Trading Strategies

Strategy	Gross	Net	Vol	Gross SR	Net SR
LmH ^I	5.20%	2.84%	1.36%	3.83	1.76
HmL ^Ω	5.78%	0.12%	2.31%	2.51	0.05
HmL ^{Γ,Ω}	6.63%	3.41%	1.54%	4.29	1.77
Market ^{end}	-1.39%	-1.39%	3.26%	-0.43	-0.43

C. Risk-Adjusted Returns

We regress strategy returns on the end-of-day market return and the standard Fama-French five factors plus momentum and intermediary capital factors. The alphas are nearly identical to the gross returns—the gamma strategy alpha is 5.21 percent (t-statistic 9.37), the ETF strategy alpha is 5.77 percent (t-statistic 6.07), and the combined strategy alpha is 6.66 percent (t-statistic 10.24). No factor loadings are economically or statistically significant except for the combined strategy’s modest positive loading on the market factor.

These alphas represent pure timing ability—the strategies earn returns only during the thirty-minute window when rebalancing flows concentrate. Over a full trading day, these strategies would generate annualized Sharpe ratios exceeding 5.0, among the largest documented in the market efficiency literature.

VII. CONCLUSION

A. Summary of Findings

This paper documents fundamental differences in how transparency shapes the market impact of non-discretionary trading flows. Leveraged ETF rebalancing, mandated and perfectly predictable, attracts accommodating liquidity that minimizes price impact and accelerates reversal. Options delta-hedging, discretionary and opaque, generates larger and more persistent price distortions that survive transaction costs and resist market learning.

These findings contribute to several literatures. First, we extend research on supply-side liquidity provision by documenting how the predictability of order flow affects the elasticity of liquidity supply. Second, we provide the first direct comparison of two economically significant sources of hedging demand, quantifying their distinct signatures in prices, spreads, and reversal dynamics. Third, we demonstrate that the fragmentation of U.S. options trading has real consequences for price efficiency by obscuring dealers’ true inventory positions.

B. Implications for Market Design

Our results have immediate implications for the ongoing policy debate regarding post-trade transparency in derivatives markets. The SEC’s current pilot program for uniform reporting of security-based swap data represents a step toward the kind of transparency that leveraged ETFs already provide. Our evidence suggests such transparency would enhance liquidity and reduce persistent price distortions.

More broadly, our findings challenge the presumption that opacity benefits market quality by protecting dealers from

predatory trading. In this setting, opacity creates the very information asymmetry that exposes dealers to adverse selection. By contrast, sunshine trading—the pre-announcement of impending large trades—benefits both the trading institution and the broader market through enhanced liquidity provision.

C. Limitations and Future Research

Several limitations suggest directions for future research. First, our gamma imbalance measure remains incomplete—we observe only five exchanges and capture only end-of-day inventory snapshots. A complete picture of dealer hedging activity would require audit trail data with millisecond timestamps. Second, we do not observe the full portfolio of dealer hedging activity, which may span multiple correlated assets and derivatives. Third, our sample period ends before the dramatic growth of zero-days-to-expiration options, whose extreme gamma profiles may amplify the effects we document.

We leave these extensions to future work. The central message of this paper is already clear: transparency is not merely a normative virtue in financial markets but a practical determinant of liquidity, efficiency, and the speed of arbitrage.

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